PDD_STAT -- User Guide

Version: 1.0

Developer: P. Demeshko

Interface language: English (labels, metrics, plot annotations)

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1. System Overview

PDD_STAT is a web application for statistical analysis of medical and clinical data. The system provides:

- 22 analysis templates (from descriptive statistics to regression models and survival analysis)
- AI assistant powered by Ollama (local) or Groq (cloud API)
- DOCX report generation (standard and AI-generated)
- Code editor for custom Python analysis
- Visualizations: matplotlib, SHAP, DCA, calibration curves

Key Features

- All output in English (metrics, tables, plot labels)
- Built-in statistical safeguards: proportional hazards testing (Schoenfeld), multicollinearity (VIF), normality tests, homogeneity of variances, calibration (Hosmer-Lemeshow)
- Multiple comparison correction (Bonferroni) in pairwise tests
- Stratified survival analysis
- Stepwise predictor selection with p-value thresholds
- DCA (Decision Curve Analysis) and calibration curves for regression models
- Bootstrap confidence intervals for metrics
- SHAP for Random Forest model interpretation

2. Installation & Startup

Requirements

Component	Version
Python	≥ 3.9
pip	≥ 21.0

Install Dependencies

```
cd app/backend
pip install -r requirements.txt
```

Start the Server

```
cd app/backend
python3 -m uvicorn app.main:app --host 127.0.0.1 --port 8000
```

Open your browser at: http://127.0.0.1:8000

The frontend requires no separate build step -- the server serves static files from app/frontend/ automatically.

3. Project & Data Management

3.1. Creating a Project and Uploading Data

1. Click "+ Create Project" (left panel).
2. Enter a project name (Latin or Cyrillic).
3. Select a data file:

- Supported formats: .xlsx, .xls, .csv

4. Click "Create".
5. After upload, the Cleaning Recommendations modal appears (see section 3.2).

Re-uploading to an existing project: click the project name with the left mouse button -- a file dialog opens.

3.2. Data Cleaning

Upon data upload, the system automatically analyzes every column and suggests a cleaning plan:

Dropped Columns

Rule	Condition	Level
HIGH_MISSING	>30% missing values	Hard
MODERATE_MISSING	20–30% missing values	Soft
CONSTANT	Singleton column	Hard
IDENTIFIER	All unique values (n>10)	Hard
TECHNICAL	Name starts with `INFO_`, `CHECK_`, `PARAMS_` etc.	Hard

Imputation

Remaining columns with missing values:

- Numeric → filled with median
- Categorical/string → filled with mode

Buttons:

- "Apply Cleaning" -- apply recommendations
- "Skip (Keep as is)" -- save raw data as-is

After cleaning, data is persisted as state/project_data.parquet (snappy compression).

3.3. Variable Management

Right panel → "Columns" tab:

- Search by variable name
- Type badge: num (numeric), cat (categorical), bin (binary), str (string)
- Actions:
- -- rename
- -- delete from dataset
- -- open Label modal:
- Chart Name -- display name on plots
- Value Labels -- label for values (e.g., 0 = No, 1 = Yes)

Quick variable selection: click a variable name to assign it to the active card field (field highlights in color).

3.4. Project Context (Description, Aim)

Click on a project item → Project Context:

- Description -- project description (used by AI for context)
- Research Aim -- study objective
- Notes -- additional notes

Context is automatically injected into the AI assistant's system prompt and AI reports.

4. Interface

4.1. Left Panel -- Project Manager

Element	Description
+ Create Project	Create a new project
Project list	Project names. Click → open/upload data. → context. → delete
AI Settings	Configure AI assistant
	Collapse/expand panel

4.2. Workspace

Central area -- container for analysis cards (see section 5).

Template menu bar (top):

Menu	Templates
**Basic Stats **	Categorical, Numeric, Correlation, Spline, Descriptive, Violin, ANOVA, Chart Builder

Menu	Templates
**Regression **	Cox, Logistic, LASSO, Random Forest
**Survival **	Kaplan-Meier
**Evaluation **	Model Evaluation (Binary), Survival Eval, ROC, Diagnostic Accuracy, Agreement
**Prediction **	Individual Prediction

Action buttons:

- AI Chat -- open AI assistant chat
- Code -- switch to code editor mode
- / -- toggle light/dark theme

4.3. Right Panel -- Variables

Tab	Content
Columns	Variable list with search, type badges, editing
Reports	DOCX report generation
Chain	Analysis history

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5. Statistical Analysis Templates

5.1. Categorical Comparison

Method: χ^2-test / Fisher's Exact Test / Monte Carlo simulation

Parameters:

Field	Description
`col1`	First categorical variable (table rows)
`col2`	Second categorical variable (table columns)

Test selection logic:

1. All expected frequencies ≥ 5 → χ^2-test
2. Low expected AND 2×2 table → Fisher's Exact Test
3. Otherwise → Monte Carlo (10,000 iterations)

Output metrics:

Metric	Interpretation
`chi2`	Chi-square statistic
`p_value`	Significance level; **p < 0.05** = significant association
`df`	Degrees of freedom
`min_expected_frequency`	Minimum expected cell count
`all_expected_ge5`	All expected ≥ 5?
`odds_ratio`	Odds ratio (Fisher, 2×2 only)

5.2. Numeric Comparison

Method: t-test (independent) / ANOVA or Mann-Whitney U / Kruskal-Wallis H

Parameters:

Field	Description
`value`	Numeric dependent variable
`group`	Categorical grouping variable

Test selection:

- If normal (Shapiro-Wilk) AND equal variances (Levene):
- 2 groups → t-test + Cohen's d
- >2 groups → ANOVA + η^2
- Otherwise:
- 2 groups → Mann-Whitney U
- >2 groups → Kruskal-Wallis H

Output metrics:

Metric	Interpretation
`statistic`	Test statistic value
`p_value`	Significance level
`cohens_d`	Cohen's d effect size (t-test only): 0.2 = small, 0.5 = medium, 0.8 = large
`normality`	Shapiro-Wilk test results per group
`equal_var`	Levene's test result

Plots: Boxplot + barplot (mean ± SD), 2 panels.

5.3. Correlation Analysis

Method: Pearson (parametric) or Spearman (rank) correlation

Parameters:

Field	Description
`predictors`	Select multiple variables
`method`	`Pearson` (assumes normality) or `Spearman` (any distribution)
`threshold`	r

Output metrics:

Metric	Interpretation
`n_variables`	Number of variables
`n_strong_pairs`	Pairs with
`strongest_r`	Maximum
`pairs`	Strong pairs with **Bonferroni-corrected p-values**. Corrects for family-wise error rate (FWER)

p-values are Bonferroni-corrected (multiplied by number of all pairwise tests) to control FWER.

Plot: Correlation heatmap (lower triangle).

5.4. Spline Analysis

Method: Linear spline analysis within Cox or Logistic regression

Parameters:

Field	Description
`variable`	Variable for spline transformation
`target`	(Logistic) Binary outcome
`time` + `event`	(Cox) Time-to-event
`covariates`	Adjustment covariates
`knots`	Knot placement: `Median`, `Quartiles`, `Custom`

Output metrics:

Metric	Interpretation
`nonlinear_p`	p-value for non-linearity test. **p < 0.05** = significant deviation from linearity
`knots`	Knot positions

Plot: Log-HR / Log-OR vs variable with knot markers + histogram.

5.5. Descriptive Statistics

Method: Summary statistics + Shapiro-Wilk normality test

Parameters:

Field/Option	Description
`covariates`	Variables (empty = all numeric)
`Include histograms`	Plot histograms per variable

Output metrics:

Metric	Interpretation
N, Mean, Median, Std, Min, Max, Q1, Q3	Standard descriptive statistics

Metric	Interpretation
Skewness	Asymmetry of distribution (0 = symmetric)
Kurtosis	Tail heaviness (3 = normal distribution)
Missing count	Number of NaN values
Outlier count	IQR method: beyond Q1-1.5×IQR / Q3+1.5×IQR
Shapiro-Wilk W, p	**p < 0.05** = distribution deviates from normality

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5.6. Violin Plots

Method: Violin plots + Mann-Whitney / Kruskal-Wallis test

Parameters:

Field	Description
`value`	Numeric variable
`group`	Grouping variable

Output metrics:

Metric	Interpretation
`mann_whitney_p`	Mann-Whitney U p-value (2 groups)
`kruskal_p`	Kruskal-Wallis H p-value (>2 groups)

Plot: Violin plot, optional box overlay and swarm points.

5.7. ANOVA / Kruskal-Wallis

Method: One-way ANOVA (parametric) or Kruskal-Wallis (non-parametric)

Parameters:

Field	Description
`value`	Numeric outcome
`group`	Categorical grouping (≥2 groups)

Output metrics (tabs):

Statistics tab:

Metric	Interpretation
`test_name`	`ANOVA` or `Kruskal-Wallis`
`F_stat` / `H_stat`	Test statistic
`p_value`	Significance level
`eta_sq`	η^2 — proportion of variance explained by group (ANOVA)
`omega_sq`	ω^2 — less biased effect size than η^2
`epsilon_sq`	ε^2 — effect size for Kruskal-Wallis
`trend_test`	Jonckheere-Terpstra trend test (≥3 groups)

Post-hoc tab:

Metric	Interpretation
Pairwise comparisons	Tukey HSD (ANOVA) or Mann-Whitney with Bonferroni (KW)
`significant`	p < 0.05 after correction

Diagnostics tab:

Metric	Interpretation
`all_normal`	All groups normal (Shapiro-Wilk p>0.05)
`equal_var`	Variances equal (Levene p>0.05)

Plot: Boxplot + jitter + barplot (mean ± SD).

5.8. Chart Builder

Method: Quick visualization (no statistical tests)

Parameters:

Field	Description

Field	Description
`Chart type`	`Histogram`, `Box Plot`, `Scatter`, `Bar (mean±SD)`, `Density`
`x`	X variable
`y`	Y variable (scatter only)
`group`	Color/group by
`Bins`	Histogram bins (default 20)
`KDE curve`	Overlay kernel density estimate (histogram)
`Regression line`	Regression line with r^2 (scatter)
`Chart Axes`	Custom title, axis labels

Plot: The requested chart type.

5.9. Cox Regression

Method: Cox Proportional Hazards Regression (lifelines CoxPHFitter)

Parameters:

Field	Description
`event`	Event indicator (0 = censored, 1 = event)
`time`	Time-to-event
`covariates`	Predictors
Covariate type	`Cat`/`Cont` toggle — categorical or continuous
`Reference groups`	Reference category for categorical variables
Univariate	Each predictor separately
Multivariate	All predictors jointly
Method	`Enter` (all at once), `Forward` (forward selection), `Backward` (backward elimination)
`p_enter`	Entry p-value threshold (default 0.05)
`p_remove`	Removal p-value threshold (default 0.10)
`Adjust Survival Curves`	Covariate for adjusted survival curves

Output metrics:

Table tab:

Metric	Interpretation
`Beta (β)`	Regression coefficient
`SE`	Standard error
`z`	z-statistic (β/SE)
HR (Hazard Ratio)	exp(β). >1 = increased hazard, <1 = decreased hazard
`95% CI`	95% confidence interval for HR
`p-value`	Wald test significance

Diagnostics tab:

Metric	Interpretation
C-index	Harrell's Concordance Index — discrimination. 0.5 = random, 1.0 = perfect. Not calibrated for censored data
AIC	Akaike Information Criterion — model quality (lower = better). Not interpretable in isolation
BIC	Bayesian Information Criterion — penalizes complexity more than AIC
Log-likelihood	Log-partial likelihood at convergence
Global LRT p	Likelihood ratio test p-value for overall model significance
Schoenfeld test	Proportional hazards assumption test. **p<0.05** = PH violation. **Critical diagnostic check**
VIF	Variance Inflation Factor. **>10** = problematic multicollinearity, **>5** = moderate

Steps tab (Forward/Backward only): stepwise selection history.

"Save Risk" button -- saves predicted risk (linear predictor) as a new column.

Plots:

- Forest plot (HR with 95% CI)
- Baseline hazard / survival function
- Martingale residuals (linearity diagnostic for first continuous predictor)
- Adjusted survival curves (if covariate selected)

Assumption checks:

- EPV (Events Per Variable) ≥ 10 -- warning if <10

- Schoenfeld -- PH test per variable and globally
- VIF -- multicollinearity
- Stepwise warning: p-values do not account for the selection process and may be inflated

5.10. Logistic Regression

Method: Logistic Regression with L2 penalty, class_weight='balanced'

Parameters:

Field	Description
`target`	Binary target variable
`predictors`	Predictors
Covariate type	`Cat`/`Cont` toggle
`Reference groups`	Reference category
Univariate	Each predictor separately
Multivariate	All predictors jointly
Method	`Enter`, `Forward`, `Backward`
Validation	`None`, `Train/Test` (80/20), `Cross-val` (5-fold CV)

Output metrics:

Table tab:

Metric	Interpretation
`Beta (β)`	Regression coefficient
OR (Odds Ratio)	exp(β). >1 = higher odds, <1 = lower odds
`95% CI`	95% confidence interval for OR
`p-value`	Wald test significance

Diagnostics tab:

Metric	Interpretation
AUC	Area Under ROC Curve — discrimination. 0.5 = random, 0.8 = excellent, 1.0 = perfect
Accuracy	(TP+TN)/(Total) — overall correct classification
Sensitivity	TP/(TP+FN) — true positive rate
Specificity	TN/(TN+FP) — true negative rate
Hosmer-Lemeshow	Calibration test: χ^2, df, p-value. **p<0.05** = significant miscalibration
VIF	Multicollinearity (>10 = problematic)
Warnings	VIF errors, convergence issues

Steps tab (Forward/Backward): stepwise selection history.

"Save Probabilities" button -- saves predicted probabilities as a new column.

Plots (3):

- ROC curve
- DCA (Decision Curve Analysis) with Treat All / Treat None baselines
- Calibration curve with Brier Score

Assumption checks:

- Target must be binary (exactly 2 unique values)
- Minimum 20 complete cases
- All predictors standardized (StandardScaler)
- Warning when validation="none": metrics are optimistic

5.11. Logistic LASSO

Method: L1-regularized logistic regression (LASSO)

Parameters:

Field	Description
`target`	Binary target
`covariates`	Predictors (empty = all numeric + encoded categorical)
`Auto-select C`	Automatic C selection via 5-fold CV

Field	Description
`C`	Inverse regularization strength (if auto-C off). Smaller C = stronger regularization

Output metrics:

Metric	Interpretation
AUC	ROC AUC (training data — **optimistic**)
Accuracy	Overall accuracy (training — **optimistic**)
`n_features_selected`	Features with non-zero coefficient
`n_features_zeroed`	Features zeroed out (removed by LASSO)
`best_C`	Best C value (if auto-select)
`selected_features`	List: feature, OR, coefficient
`intercept`	Model intercept

Plots (4):

- Coefficient bar plot (selected + top 30 all features)
- CV curve (if auto-select C)
- DCA (Decision Curve Analysis)
- Calibration curve

Important:

- Metrics are on training data -- optimistic for generalization
- class_weight='balanced' -- handles class imbalance
- solver='saga' -- appropriate for L1 penalty

5.12. Random Forest

Method: Random Forest Classifier (sklearn, 200 trees, max_depth=10, min_samples_leaf=5, class_weight='balanced')

Parameters:

Field	Description
`target`	Target variable (binary or multiclass)
`exclusions`	Columns to exclude (comma-separated)
`Top features`	10, 15 (default), 20, All
`SHAP analysis`	Enable SHAP calculation (may be slow)

Output metrics:

Metric	Interpretation
OOB score	Out-of-Bag R^2 — unbiased performance estimate from unused trees
Accuracy	Test set accuracy (75/25 split)
AUC	ROC AUC (binary targets only)
Precision	PPV = TP/(TP+FP)
Recall	Sensitivity = TP/(TP+FN)
F1	Harmonic mean of precision and recall
`top_features`	Gini importance — contribution of each feature to node purity. Not a p-value!
`shap_features`	Mean absolute SHAP — more interpretable feature contribution magnitude

Plots:

- Feature importance bar (Gini)
- SHAP bar + summary plots (if enabled)

Post-run buttons:

- "Save Model" -- saves model as .joblib (with metadata: features, target, metrics)
- "Save Probabilities" -- saves predicted probabilities as new column

Note: Auto-excludes columns matching: _event, event_, _time, time_, predicted_, _prob, censored, followup, cox_, logistic_. Only numeric predictors are used.

5.13. Kaplan-Meier

Method: Kaplan-Meier survival curves / Nelson-Aalen cumulative hazard

Parameters:

Field	Description
`time`	Time-to-event
`event`	Event indicator
`group`	Grouping variable
`stratify`	Stratification variable (separate analysis per stratum)
`Survival` / `Hazard`	Plot type: survival probability or cumulative hazard
`95% CI`	Show confidence intervals
`Chart Axes`	X step (months), Y step (%), labels

Output metrics:

Metric	Interpretation
`summary`	Per group: n, events, median survival (months)
Log-rank p	Overall log-rank test p-value. **p<0.05** = groups differ significantly
Pairwise log-rank	Bonferroni-corrected pairwise comparisons. Shows which specific groups differ
`number_at_risk`	Number-at-risk table at key time points
`survival_probability`	S(t) survival probability table
`median_survival`	Median survival per group. `NR` = not reached

Plot: Kaplan-Meier survival curve S(t) or Nelson-Aalen cumulative hazard H(t).

5.14. ROC Analysis

Method: ROC analysis for multiple numeric predictors vs binary target

Parameters:

Field	Description
`target`	Binary target
`Event value`	Value considered "event" (1 or 0)
`predictors`	Numeric predictors

Output metrics:

ROC Results tab:

Metric	Interpretation
AUC (95% CI)	Area Under ROC Curve. Hanley-McNeil CI method
`Threshold`	Optimal cutpoint via Youden's J (sens + spec - 1)
`Sensitivity`	True positive rate at optimal threshold
`Specificity`	True negative rate at optimal threshold
PPV	Positive predictive value
NPV	Negative predictive value

Comparison tab (≥2 predictors):

Metric	Interpretation
DeLong test	Pairwise AUC comparison. **p<0.05** = AUCs differ significantly

Plot: Combined ROC curves for all predictors.

5.15. Model Evaluation (Binary)

Method: Comprehensive binary classification evaluation with bootstrap CI

Parameters:

Field	Description
`target`	Binary target
`predictors`	Predicted probability columns (0-1)
`Bootstrap samples`	500, 1000 (default), 2000
`DCA`	Include Decision Curve Analysis
`Calibration`	Include calibration curves

Output metrics:

Metric	Interpretation

Metric	Interpretation
AUC (bootstrap CI)	ROC AUC with bootstrap 95% CI
`Accuracy`	Overall classification accuracy
`Sensitivity` / `Specificity`	True positive and negative rates
PPV / **NPV**	Predictive values
F1	Harmonic mean of precision and recall
Brier Score	Mean squared probability error. 0 = perfect, <0.25 = good, >0.25 = poor
DCA Net Benefit	Clinical net benefit at thresholds 5-50%. NB = (TP/n) - (FP/n) × (p/(1-p))

Plots:

- DCA curves (all models on one plot)
- Calibration curves (one per model)

5.16. Survival Prediction Evaluation

Method: Survival model evaluation: C-index, time-dependent AUC, bootstrap CI

Parameters:

Field	Description
`time`	Time-to-event
`event`	Event indicator
`predictors`	Risk score/prediction columns
`Evaluation time points`	Time points for AUC (comma-separated, e.g., "12,24,48")
`stratify`	Stratification variable
`Bootstrap samples`	500, 1000 (default), 2000
`Smooth`	Smoothed AUC curves

Output metrics:

Metric	Interpretation
C-index (bootstrap CI)	Harrell's C-index with bootstrap 95% CI. 0.5 = random
Time-dependent AUC	Cumulative dynamic AUC as function of time
ΔC (model comparison)	C-index difference between models with bootstrap p-value
Strata	Separate results per stratum level

Plot: Time-dependent AUC (optionally smoothed).

5.17. Diagnostic Accuracy

Method: Diagnostic accuracy metrics for binary tests vs gold standard

Parameters:

Field	Description
`target`	Reference standard (binary, "gold standard")
`predictors`	Index tests (binary or probability 0-1)

Output metrics (per test):

Metric	Interpretation
Sensitivity (95% CI)	TP/(TP+FN). Proportion of correctly identified positives
Specificity (95% CI)	TN/(TN+FP). Proportion of correctly identified negatives
PPV (95% CI)	TP/(TP+FP). Probability that a positive test reflects true disease
NPV (95% CI)	TN/(TN+FN). Probability that a negative test reflects true absence
Accuracy (95% CI)	(TP+TN)/(Total). Overall correct classification
LR+ (95% CI)	Positive likelihood ratio = Sens/(1-Spec). >10 = strong evidence
LR- (95% CI)	Negative likelihood ratio = (1-Sens)/Spec. <0.1 = strong evidence
DOR	Diagnostic Odds Ratio = (TP×TN)/(FP×FN). 1 = useless test

Comparison tab (>1 test):

Metric	Interpretation
McNemar test	Paired comparison. **p<0.05** = tests differ significantly

Contingency tables tab: TP/FP/FN/TN per test.

Plot: Forest plot of sensitivity and specificity with 95% CI.
Note: Probability columns (0-1) are auto-thresholded at 0.5.

5.18. Agreement Analysis

Method: Cohen's Kappa (2 raters) / Weighted Kappa (ordinal) / Fleiss' Kappa (3+ raters)

Parameters:

Field	Description
`predictors`	Rater columns (≥2 required)
`Nominal` / `Ordinal`	Scale type
`Linear` / `Quadratic`	Weight type (ordinal only)

Output metrics:

Metric	Interpretation
Kappa	Agreement coefficient corrected for chance. Range: -1 to 1
`95% CI`	Confidence interval
`p-value`	Significance vs zero
Interpretation	Landis & Koch: Poor (≤0), Slight (0.01-0.20), Fair (0.21-0.40), Moderate (0.41-0.60), Substantial (0.61-0.80), Alm
`percent_agreement`	Raw agreement percentage

For ≥3 raters: Fleiss' Kappa with per-category breakdown.

Plot: Confusion matrix heatmap (2 raters).

5.19. Individual Prediction

Method: Single-patient prediction from saved model with SHAP explanation

Parameters:

Field	Description
Select File	Upload .joblib or .pkl model
Patient form	Radio buttons for categorical features, number inputs for numeric

Output metrics:

Metric	Interpretation
`prediction`	Predicted class
`probability`	Probability of class 1

Plot: SHAP waterfall plot -- contribution of each feature to the prediction.

6. AI Chat

6.1. AI Settings

Click "AI Settings" (left panel or above AI Chat):

Parameter	Description
Provider	`Ollama (Local)` — run models locally. `Groq (Cloud)` — cloud API
Ollama URL	Ollama server address (default `http://localhost:11434`)
Ollama Model	Model (loaded after "Test Connection")
Groq API Key	API key (get at console.groq.com)
Groq Model	Model (list loaded after "Test Connection")
Temperature	Generation temperature: 0 = deterministic, 1 = creative
Max Tokens	Maximum response length (100-8000)
System Prompt	AI instruction prompt (includes project context, dataset info, analysis history)

Buttons:

- "Test Connection" -- test provider connectivity and load model list
- "Connect" -- confirm provider selection
- "Reset to default" -- reset system prompt to default

- Save / Cancel -- save or discard settings

6.2. Modes: Assistant and Coder

AI Assistant (default):

- Conversational chat with AI
- Answers statistical questions, interprets results
- Has context of project description, dataset, and analysis history
- Supports markdown formatting

AI Coder ("Coder" toggle):

- Generates Python analysis code
- Auto-extracts `python code blocks and executes them
- Shows code output and generated plots
- Self-corrects on errors (up to 3 retries)
- Uses temperature 0 (deterministic output)

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7. Reports

Right panel → Reports tab.

7.1. Standard DOCX Report

Parameter	Description
Analyses	`All analyses` or `Selected` (with checkboxes)
Sections	Dataset overview, Results tables, Key metrics
Title	Report title
Generate DOCX	Create report. Download via link

Contents:

- Dataset overview (row count, column types)
- For each analysis: metrics, tables from preview text

7.2. AI Report

Parameter	Description
Language	Report language (user prompt, default "English")
Generate AI Report	Create AI-generated report

Contents (AI-generated):

1. Objective -- based on project context
2. Materials -- dataset description + basic stats
3. Methods -- statistical methods used
4. Results -- key findings with exact numbers (HR, OR, p-values, AUC, C-index)
5. Conclusion -- brief summary

Requirements:

- AI must be configured (valid provider and credentials)
- At least one analysis in history (max 5 included in prompt)

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8. Code Editor Mode

Click "Code" in the header.

Action	Description
Type Python code in the monospace text area	
Run	Execute via `exec()` on the server
Clear Code	Clear editor
Clear History & Plots	Delete all analysis history and plot files
Exit Code Mode	Return to analysis view

Available variables in code:

- df -- project pandas DataFrame
- pd, np, plt -- standard libraries
- save_plot(name) -- save plot with auto-close
- fmt_p(p) -- format p-value (shows <0.0001 if tiny)
- get_label(var, value) -- variable/value labels

9. Analysis History (Chain)

Right panel → Chain tab.

- Chronological list of completed analyses (reverse order)
- Color coding:
- Blue -- regression
- Green -- survival
- Amber -- evaluation
- Gray -- basic tests
- Purple -- Random Forest
- Click an item to restore the analysis card with results

10. Tips & Best Practices

Action	Tip
Enter in AI input	Send message
Shift+Enter	New line
Click a variable	Assign to active field
Hide Charts	Free up memory by hiding plots
** / **	Toggle theme

Workflow recommendations

1. Start with Descriptive Statistics -- inspect distributions, missing values, outliers
2. Use Categorical + Numeric Comparison for preliminary screening
3. For regression: start with Univariate, then Multivariate (Enter), then Stepwise if needed
4. Always check Schoenfeld PH test in Cox regression -- this is the key assumption
5. VIF > 10 -- remove collinear predictors and re-run
6. DCA helps assess clinical utility, not just statistical significance
7. Bootstrap CI provides more realistic metric precision estimates
8. Don't rely solely on Stepwise p-values -- they don't account for the selection process
9. LASSO with auto-select C is a good choice for high-dimensional feature selection
10. AI Coder mode is useful for quick ad-hoc analyses without leaving the app